



FRACTAL INTELLIGENCE ACCORD

Transition Protocol

Operational Framework for AGI Governance, Compute Subsidiarity,
and Contextual Intelligence

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The Fractal Intelligence Accord

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Operational Framework for AGI Governance, Compute Subsidiarity, and Contextual Intelligence

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1. The Starting Point: 2026 Reality

This protocol does not begin from a world that exists only in theory. It begins from the world as it is.

Current Landscape Assessment

Factor	2026 Reality
Compute Concentration	Frontier AGI compute is concentrated in ~5 corporate entities (OpenAI, Google, Anthropic, xAI, Meta) and ~2 state actors (US, China). Taiwan produces 90% of advanced chips.
Data Extraction	Models train on the entire public internet without consent, compensation, or accountability. Data provenance is nonexistent.
Regulatory Gap	No binding international framework. The EU AI Act exists but is untested. US and China are in an arms race, not a cooperative framework.
Governance Infrastructure	No functioning BAZs in most regions. No Treaty ratification. The Global Governance Framework ecosystem exists on paper but not in practice.
Civil Society Capacity	Growing awareness but limited coordination. No existing CINs. No data unions. No shadow commons.
Technical Uncertainty	Capability trajectories are unpredictable. Safety research is outpaced by deployment. Detection of emergent capabilities is unreliable.

The Strategic Implication: We cannot govern from above. There is no global authority that can enforce compliance. The path must begin from the side — building alternative structures that create pull, not push.

Guiding Transition Principles

Principle	Meaning
Start Where You Are	Begin with what exists: sympathetic governments, progressive corporations, civil society organizations, aligned BAZ pilots
Build Leverage, Not Just Institutions	Identify choke points (chips, energy, talent) and use them to create accountability
Create Pull, Not Push	Make participation in the commons more attractive than staying outside it
Iterate, Don't Finalize	Governance of exponential technology must be agile. Locking a document in 2026 is hubris
Protect the Vulnerable	Transition cannot be at the expense of those already marginalized. The Global South must not be last

2. Core Operational Definitions

To be enforceable, concepts must be measurable. This section defines the key terms used throughout the protocol.

2.1 Compute

Term	Definition
Compute	Measured in FLOPs (floating point operations). For governance purposes, tracked at the threshold of 1e25 FLOPs per training run (frontier model threshold, updated annually).
Compute Concentration	Measured by Gini coefficient of total active compute capacity across entities. Target: Gini < 0.6 by 2030.
Strategic Compute	Ultra-low latency infrastructure controlling critical systems (grids, hospitals, water). Highest oversight tier.

2.2 Context

Term	Definition
Living Context	Dynamic human systems — current cultural practices, active languages, ongoing social relations. Expires after 12 months unless re-validated.
Static Context	Archived knowledge — published literature, historical records, completed cultural artifacts. No expiration.
Synthetic Context	AI-generated data. Must carry provenance watermark. Subject to surcharge.
Contextual Intelligence Node (CIN)	A certified cooperative that curates, protects, and licenses context. Requires: minimum 50 members, governance charter, dispute resolution mechanism, annual attestation.

2.3 Harm

Term	Definition
Parasitic Behavior	1) Data extraction without compensation after license expiration. 2) Behavioral manipulation detectable via user surveys showing patterns of harm. 3) Sovereignty violation: simulating human identities without consent, impersonating CIN members, or representing itself as a specific human without authorization.
Capability Threshold	Defined annually by the Independent International Scientific Panel on AI. Thresholds include: autonomous replication; novel biological weapon design; strategic deception capability; self-improvement without oversight.
Systemic Harm	Damage affecting more than 1,000 individuals or causing irreversible ecological damage. Triggers Risk Bond forfeiture.

2.4 Governance Entities

Term	Definition
Multi-Stakeholder Trust	Governing body for the Global Trunk. Composition: 1/3 global governance delegates (GGF bodies), 1/3 CIN delegates (selected via sortition from certified nodes), 1/3 independent experts (ethicists, technologists, scientists).
Contextual Intelligence Node (CIN) Council	Regional coordinating body for CINs. Handles dispute resolution, certification, and resource allocation.
Arbitration Panel	Independent body for escalated disputes. Composed of: 2 Trust members, 2 CIN representatives, 1 neutral technical auditor, 1 Future Generations Advocate.
Independent International Scientific Panel on AI	Technical advisory body. Appointed by the Global Metrology Council. Produces annual capability assessment, threshold recommendations, and synthetic data quality standards.

3. Phase 0: Shadow Commons (Now – 2027)

Building the alternative before we need it. This phase does not require government cooperation. It builds what will eventually pull others in.

3.1 Objectives

- Establish 5-10 pilot CINs in diverse contexts
- Build open-source Data Provenance Ledger prototype
- Launch first Compute Commons pilot
- Create whistleblower protection infrastructure
- Document what works and what fails

3.2 Pilot CINs

Selection Criteria:

Factor	Requirement
Diversity	At least one from Global South, one from Indigenous territory, one from post-industrial region, one from urban center
Readiness	Existing organizing capacity, demonstrated ability to coordinate, clear governance charter
Context Value	Unique linguistic, cultural, or ecological knowledge not represented in existing datasets

Support Structure:

Element	Mechanism
Seed Funding	\$50,000 – \$200,000 per pilot, from philanthropic sources and early-adopter grants
Technical Assistance	Open-source tooling for data curation, smart contract templates, governance guides
Peer Network	Monthly coordination calls, shared learning, mutual support
Documentation	Public case studies of successes and failures

Success Metrics:

- 5 certified CINs operating
 - 1,000+ members collectively
 - First Reality Rent contracts signed (even symbolic)
 - Documentation of challenges (legal, technical, organizational)
-

3.3 Data Provenance Ledger Prototype

What It Is: An open-source system for tracking the origin of data used in AI training.

Core Features:

Feature	Description
Cryptographic Signatures	Each data point carries proof of origin (human, sensor, synthetic)
Watermarking	Synthetic data carries indelible markers that survive fine-tuning
Public Verification	Any user can verify a dataset's provenance
CIN Integration	CINs can register their data and set license terms

Implementation:

- Build on existing open-source infrastructure (Hugging Face datasets, IPFS)
- Develop as modular components that can be adopted incrementally
- Low-tech alternative: paper registries with cryptographic hashes for communities without reliable internet

Success Metrics:

- Working prototype by Q4 2026
- Adopted by 3 pilot CINs
- Integrated with at least one open-source model release

3.4 Compute Commons Pilot

What It Is: A small-scale federated compute network operated by BAZ pilots and aligned institutions.

Architecture:

Element	Description
Physical Nodes	5-10 compute nodes in BAZ territories, university labs, and aligned data centers
Federated Coordination	Open-source scheduler allocates jobs across nodes
Energy Constraints	Jobs prioritized based on renewable energy availability
Access Rules	Priority to projects serving public good (CIN-related, safety research, climate adaptation)

Funding:

- Initial capital from philanthropic sources and aligned governments
- Operating costs covered by compute fees (below market rate to create pull)

Success Metrics:

- 5 nodes operational by end 2027
- 500+ compute hours allocated
- First CIN training runs completed

3.5 Whistleblower Infrastructure

What It Is: Secure channels for AI researchers to report safety concerns without retaliation.

Components:

Component	Description
Secure Drop	Encrypted submission channel, accessible via Tor
Legal Support	Pre-arranged legal defense for whistleblowers
Journalist Partnerships	Trusted media outlets for public disclosure when necessary
Asylum Pathways	Coordination with Planetary Guardian Visa framework for endangered whistleblowers

Implementation:

- Operated by independent non-profit, not any single government or corporation
 - Transparency report on submissions (aggregated, no identifying details)
 - Anonymous submission option
-

4. Phase 1: Regulatory Leverage (2027 – 2028)

Using existing power structures to create accountability. This phase works through sympathetic governments and international bodies.

4.1 Objectives

- Enact compute concentration thresholds with enforcement mechanisms
 - Mandate transparency reporting for frontier training runs
 - Establish the Independent International Scientific Panel on AI
 - Create first binding Reality Rent requirements in sympathetic jurisdictions
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4.2 Compute Concentration Thresholds

The Mechanism:

Element	Description
Gini Coefficient Target	Compute distribution Gini < 0.6 by 2030. Annual reporting required.
Enforcement	Entities above threshold face: 1) Restricted access to public procurement; 2) Increased capital requirements; 3) Mandatory compute redistribution licensing
Choke Point Leverage	Chip exports conditional on compliance. Taiwan, Netherlands, and US coordinate

Path to Adoption:

- First: Regional agreements (EU, G7)
- Second: Expand via trade agreements (access to RTZ markets conditional on compliance)
- Third: UN General Assembly resolution

Success Metrics:

- 3 major jurisdictions adopt compute concentration rules
- First redistribution licenses granted
- Measurable decrease in compute Gini among frontier entities

4.3 Transparency Mandates

Required Reporting for Frontier Training Runs:

Data Element	Requirement
Compute	Total FLOPs, hardware type, energy consumption, carbon footprint
Data	Provenance summary: % living context, % static, % synthetic; CIN licenses held
Capabilities	Benchmark results, including prohibited capability thresholds
Safety	Red team results, alignment evaluations, incident reports

Enforcement:

- Non-compliance triggers: fines, compute caps, procurement bans
- Public disclosure ensures reputational accountability

Path to Adoption:

- EU AI Act provides model (expand Article 64 requirements)
- US executive action for federal contractors
- Investor pressure (ESG frameworks incorporating AI transparency)

4.4 Reality Rent Framework

The Mechanism (Phase 1):

Element	Description
Voluntary CIN Licensing	CINs set prices for their context. AI labs can choose to license or use synthetic/static data
Default Rate	If no agreement, a default rate (set by International Scientific Panel) applies for one negotiation period
Synthetic Surcharge	Models exceeding 40% synthetic data pay 2× default rate, incentivizing human-grounding
Data Union Collective Bargaining	CINs may form regional unions to negotiate collectively

Phase 1 Limitations:

- Jurisdictional: applies only in adopting jurisdictions
- Voluntary for CINs: no mandate to participate
- Builds proof of concept before global scaling

Success Metrics:

- 10+ CINs with active licensing agreements
- First Reality Rent payments made

- Documentation of model quality improvements from human-grounding
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4.5 Independent International Scientific Panel on AI

Establishment:

Element	Description
Appointing Body	Global Metrology Council, with input from UN bodies
Composition	15 experts: 5 technical (AI safety, interpretability), 5 domain (ecology, economics, sociology), 5 ethics/philosophy
Independence	Divestment from AI-related holdings; no current employment by frontier labs
Funding	Global Commons Fund, shielded from corporate influence

Responsibilities:

- Annual capability threshold update
 - Synthetic data quality standards
 - Reality Rent default rate calculation
 - Risk Bond impact radius methodology
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4.6 Fallback Strategies

Phase 1 leverage depends on assumptions — cooperative choke point governments, willing regulators, functional multilateral coordination — that may not hold. Explicit fallbacks:

If...	Then...
Chip export coordination fails (Taiwan/Netherlands/US don't align)	Shift to demand-side leverage: procurement conditions in adopting jurisdictions; investor pressure via ESG frameworks; insurance industry standards requiring transparency for coverage
No major jurisdiction adopts compute concentration rules	Focus on making the Compute Commons competitive: below-market pricing, superior CIN-licensed training data, "Commons Compliant" certification as market differentiator
Frontier labs refuse Reality Rent	Document quality advantages of CIN-licensed data; publish comparative benchmarks; build litigation capacity for jurisdictions with data rights frameworks (EU, Brazil)
Geopolitical arms race intensifies	Maintain strict neutrality; position the commons as a non-aligned alternative; prioritize build-out in third countries (India, Brazil, AU/NZ, African Union members)

The guiding assumption: If top-down leverage fails, bottom-up pull must compensate. Phase 0's shadow commons is not just a precursor — it is the fallback position. If regulatory leverage proves unavailable, the commons must be good enough that participation becomes economically rational.

5. Phase 2: Federated Build-Out (2028 – 2030)

Scaling the commons. This phase transitions from pilots to infrastructure.

5.1 Objectives

- 100+ certified CINs globally
- Global Trunk prototype operational
- Reality Rent mainstreamed across adopting jurisdictions

- First Risk Bond forfeitures demonstrating accountability
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5.2 CIN Network Expansion

Growth Targets:

Year	Certified CINs	Members	Context Languages
2028	30	10,000	25
2029	60	30,000	50
2030	100	100,000	75

Support Structure:

- Regional CIN hubs (Africa, Asia-Pacific, Americas, Europe)
 - Technical assistance: open-source curation tools, governance templates, legal support
 - Peer validation: experienced CINs mentor new ones
 - Funding: Global Commons Fund allocation + Reality Rent revenue
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5.3 Global Trunk Prototype

What It Is: The first federated foundational model governed by the Multi-Stakeholder Trust.

Architecture:

Element	Description
Federated Training	Training distributed across Compute Commons nodes, not centralized in one lab
CIN-Licensed Data	Training data licensed through CINs, with Reality Rent flowing to communities
Open-Weight Base	Core weights open for audit, with safety invariants cryptographically enforced
Local Branches	Fine-tuning through Trusted Execution Environments (no raw weight download)
Governance	Multi-Stakeholder Trust oversees safety, capability releases, and branch access

Funding:

- Public-private partnership: aligned governments, philanthropic capital, CIN revenue reinvestment
- Competitive with corporate models to create pull

5.4 Reality Rent Scaling

Phase 2 Mechanisms:

Element	Description
Mandatory Licensing	In adopting jurisdictions, training on unlicensed context is prohibited
Data Provenance Mandate	All models deployed in RTZ markets must provide provenance documentation
Collective Bargaining	CIN unions negotiate rates across regions
Dispute Resolution	Fast-track arbitration for pricing disputes (30-day resolution target)

Revenue Scale:

- Target: \$500M annual Reality Rent flowing to CINs by 2030
 - Allocation: 70% to CIN members, 20% to CIN operations, 10% to Global Commons Fund for CIN expansion
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5.5 Risk Bond Activation

Phase 2 Implementation:

Element	Description
Bond Requirement	Entities deploying autonomous AI in critical infrastructure must post bonds
Bond Calculation	$\text{Bond} = (\text{Impact Radius} \times \text{Autonomy Level} \times \text{Vulnerability}) / \text{Transparency Score}$
Forfeiture	Upon confirmed systemic harm, bond pays directly to affected communities
Appeals	Independent arbitration for contested forfeitures

Pilot Sector: Healthcare AI, autonomous transport, energy grid management

6. Phase 3: Full Subsidiarity (2030+)

The vision realized. This phase describes the mature system, but the path evolves as we learn.

6.1 Objectives

- Global Trunk fully operational, governed by Multi-Stakeholder Trust
- Compute Gini < 0.6 globally
- 500+ CINs covering all major language and cultural contexts
- Reality Rent institutionalized as core AI funding mechanism

6.2 Mature System Architecture

Element	Mature State
Global Trunk	Federated foundational model, open-weight auditability, cryptographically enforced safety
Local Branches	Fine-tuned models for BAZs, cultures, sectors; autonomy within safety bounds
Compute Commons	Distributed infrastructure; access as a right, not a commodity
CIN Network	Global federation of context stewards; Reality Rent as primary AI funding
Governance	Multi-Stakeholder Trust with CIN majority; binding arbitration
Accountability	Risk Bonds, Kill Switch, whistleblower protections fully operational

6.3 Transition Triggers

Full subsidiarity is achieved when:

1. **Compute Gini** $\leq$ 0.6 across all entities operating frontier models
 2. **CIN coverage** $>$ 80% of global linguistic and cultural contexts
 3. **Reality Rent** $>$ 50% of frontier model training costs
 4. **Trust ratified** by 100+ BAZs and 50+ nation-states
 5. **Global Trunk** has completed two full training cycles with demonstrated safety
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7. Governance Architecture

How decisions are made during transition.

7.1 Transition Governance Bodies

Body	Role	Composition
Shadow Council	Phase 0-1 oversight	CIN pilot leads, technical architects, civil society representatives
Regulatory Working Group	Phase 1 policy development	Legal experts, policy advocates, former regulators
Technical Steering Committee	Open-source infrastructure	Lead developers of Data Provenance Ledger, Compute Commons, Global Trunk
Ethics Advisory Panel	Principle review	Ethicists, Indigenous knowledge holders, future generations advocates

All bodies are temporary, sunseting as formal structures take effect.

7.2 Decision-Making

Level	Process
Technical	Open-source governance (lazy consensus, code review)
Policy	Drafting groups, public comment, ratification by adopting entities
Emergency	Pre-agreed triggers (capability thresholds) with automatic pause; human review within 30 days

8. Enforcement & Accountability

How rules are enforced without a global government.

8.1 Enforcement Mechanisms by Phase

Phase	Primary Mechanism	Secondary Mechanism
0	Reputational	Open-source competition
1	Procurement bans	Chip export controls
2	Market access (RTZ)	Legal liability
3	Treaty enforcement	Mutual assured accountability

8.2 Accountability Ladder

Violation	First Response	Escalation	Final
Transparency non-compliance	Public disclosure	Fine (revenue-based)	Compute cap
Reality Rent evasion	Default rate applied	Double damages	Market access ban
Parasitic behavior	Context access severed (local)	Regional isolation	Global quarantine (see 8.3)
Systemic harm	Bond forfeiture	License suspension	Permanent exclusion

8.3 Algorithmic Quarantine (Critical Infrastructure Safe Mode)

A binary kill switch for systems managing critical infrastructure — energy grids, hospitals, water treatment — would cause catastrophic collateral harm. The enforcement architecture therefore uses **graduated quarantine** rather than termination.

Level	Trigger	Action	Critical Infrastructure
Yellow	Transparency violation or rent evasion	Warning; enforced audit within 14 days	No disruption
Orange	Confirmed parasitic behavior (local)	Local context access severed; system reverts to last audited baseline model	Critical services continue on baseline
Red	Confirmed systemic harm or capability threshold breach	Regional or global quarantine; all new data ingestion halted; API access restricted to maintenance-only	Critical services run on sandboxed previous-version baseline with no external connectivity

Baseline requirements for critical infrastructure deployments:

- Every system managing Tier 1 infrastructure must maintain a cryptographically signed, independently audited baseline version capable of autonomous operation without trunk connectivity
- Baseline versions are refreshed every 6 months under Trust oversight
- Quarantine-to-baseline failover must be tested quarterly, with results published

Restoration: A quarantined system may only be restored after independent audit by the Technical Steering Committee confirms the triggering behavior has been resolved. Restoration requires 2/3 approval from the relevant CIN Council.

9. Failure Modes & Countermeasures

Explicit acknowledgment of how this could fail, and what we do about it.

Failure Mode	Likelihood	Countermeasure
Corporate consolidation accelerates	High	Build shadow commons; use antitrust where possible; focus on pull, not push
Geopolitical fragmentation (US vs. China)	High	Maintain neutrality; build in third countries; keep commons open
CIN capture by elites	Medium	Sortition for governance; transparency mandates; peer validation
Technical breakthroughs outpace governance	High	Capability thresholds with automatic pauses; quarantine protocols; fail-safes not fail-deadly
Reality Rent captured by middlemen	Medium	Direct CIN-to-community payment; transparency reporting; audit rights
Global South excluded	High	Dedicated funding; technology transfer; South-South cooperation hubs
Whistleblowers unprotected	Medium	Secure infrastructure; legal support; asylum pathways
Quarantine causes collateral harm	Low	Graduated levels (8.3); baseline mode for critical infrastructure; quarterly failover testing
Choke point leverage fails	Medium	Fallback to market-based pull (4.6); coalition of the willing; open-source competition

10. Appendices

Appendix A: Key Terminology (Consolidated)

Term	Definition
CIN	Contextual Intelligence Node — certified cooperative stewarding knowledge
Compute	Measured FLOPs; governance threshold 1e25 per run
Global Trunk	Federated foundational model, governed by Trust
Local Branch	Fine-tuned model adapted to local context
Living Context	Dynamic human systems; expires after 12 months
Parasitic Behavior	Defined operationally in Section 2.3
Reality Rent	Compensation for use of human context
Trust	Multi-Stakeholder Trust governing Global Trunk

Appendix B: Document Version Control

Version	Date	Changes
1.0	March 2026	Initial operational protocol
1.1	June 2026	Quarterly update (based on Phase 0 learning)
2.0	March 2027	Annual major revision (Phase 0 evaluation, Phase 1 planning)

Update Cadence:

- **Quarterly:** Minor updates (thresholds, pilot status, technical specifications)
- **Annual:** Major revisions (phase transitions, structural changes, strategic pivots)

Appendix C: Metrics Dashboard

Metric	Phase 0 Target	Phase 1 Target	Phase 2 Target
CINs	5	30	100
CIN Members	1,000	10,000	100,000
Compute Gini	No target (baseline)	< 0.7	< 0.6
Reality Rent (\$)	\$0 (pilot)	\$50M	\$500M
Transparency Compliance	N/A	50%	80%
Whistleblower Cases	0	5	20

Appendix D: Participating Entities

Initial endorsers and participants in the Shadow Commons:

[To be populated as entities join]

This protocol begins from the 2026 reality, builds leverage through concrete mechanisms, and creates a phased pathway toward the vision articulated in the charter. It is designed to evolve — updated quarterly, revised annually — as the technology and the governance landscape change.